TENDENCY OF UTERINE NEOPLASM MORTALITY IN WOMEN IN SOUTHERN BRAZIL

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Abstract

The aim of this study was to analyze the trend in mortality from uterine neoplasms and their risk factors, specified by the International Classification of Diseases - 10th Edition (ICD-10): C53, C54 and C55. This is an ecological study of a cross-sectional cohort and time series of deaths from uterine cancer that were recorded on the death certificates of 147 women residing in Maringá-PR from 2004 to 2014. Deaths attributed to malignant neoplasms of the cervix, unspecified (C53.9) were the most common among these women. There was an increasing trend in mortality rates from cancer of the uterus. The behavior of age groups showed an increase in women over 65 years of age. Overall, 71.5% of the women had less than 8 years of education, and 60.5% had no partner. It was also found that the health unit in Vardelina had the highest number of deaths, corresponding to 0.73% of the population. The percentage of the total deaths in the city was 0.15%. These data indicate that women in lower social strata have less access to health services and less information regarding primary prevention and the effective treatment of uterine cancer.

Keywords: Mortality; Integral Assistance to Women's Health; Uterine neoplasms

Introduction

Cervical cancer neoplasms rank third among the most common cancers affecting women, especially in low- and middle-income countries (LMIC) such as Brazil [1]. In the Americas, mortality attributed to cervical cancer has been estimated to be 38,362 deaths in 2015. Together with other uterine neoplasms, the number of deaths reached 52,215. Morbidities were reported in an alarming number of women (88,492). In Brazil, the incidence of cervical cancer was 18.20 per 100,000 women between the ages of 50-65 years, and approximately 16,340 new cases were observed from 2016 [2].

Despite this high incidence, uterine neoplasms consist of malignant cervical neoplasms, malignant neoplasms of the uterus, and malignant uterine neoplasms of unspecified origin [1]. Neoplasms of the uterus, specifically the endometrium, are the sixth most common of all cancers [3]. The incidence is greater (5.5%) in high-income countries (HIC) compared to low-income countries (LIC) (4.2%), although mortality due to uterine neoplasms is higher in less developed countries. The cumulative risk of endometrial cancer affecting women aged 75 is estimated to be 1.6% in high-income countries and 0.7% in LIC [4]. Endometrial cancer is a neoplasm that primarily affects perimenopausal and postmenopausal women, given that in nearly 90% of cases, abnormal bleeding occurs in the early stages of the disease. The definitive diagnosis of endometrial neoplasia is established through a histological examination of material obtained through an endometrial biopsy, curettage, or hysteroscopy [5].

The occurrence of uterine cancer is strongly associated with precarious social living situations, low human development, absent or fragile community education infrastructure and difficulties accessing public health services for the early diagnosis and treatment of premalignant lesions [6]. Estimates made by the Economic Policy Department for the concentration of income among the richest in Brazil point out that there is high inequality at the top of income distribution, and it tends to limit equal opportunities in society, which can inhibit economic growth. The richest 1% of the Brazilian population has 48.5% of the income of the richest 5%. The level is comparable to that of Germany (49.4%) and the United States (51.5%), which are the top developed countries when inequality is measured in this way [7]

However, studies of data on mortality trends due to uterine neoplasms that address the cervix (C53), uterus (C54) and unspecified regions (C55) are currently scarce. It is important to highlight that the effective detection of premalignant neoplasms is related to a reduction in mortality and health care costs, which may offer benefits to population health and economic spheres that are useful in evaluating the cost-effectiveness of the preventative efforts of regular screening [8].

Thus, the objective of this study was to analyze mortality trends due to uterine neoplasms and associated risk factors utilizing the criteria of the 10th revision of the International Statistical Classification of Diseases and Related Health problems (ICD-10): C53, C54, C55 in women in southern Brazil".

Materials and Methods

Study Design and Setting

This was an ecological cross-sectional design incorporating time series methods to study the mortality associated with uterine neoplasms, specifically, C53, C54, C55 of the ICD-10, utilizing mortality records of female residents in the city of Maringá in Brazil over a ten-year period from 2004-2014.

Maringá, a medium-sized city in Southern Brazil, has a population of 397,437 inhabitants, of whom 185,353 (46.64%) are female, and 98,625 (53.21% of females) are older than 20 years. Maringá has a high level of human development (.808), the 23rd highest of all Brazilian municipalities [9]. In relation to primary care services, the Family Health Strategy uses 72 family health teams operating out of 32 health centers to reach 240,156 people, corresponding to a 67.3% coverage level. It is divided into 9 Nuclei for the Support of Family Health (NSFH) and 556 census tracts [10].

Data Sources

Cases of uterine neoplasm mortality were selected according to the ICD-10 from secondary databases of epidemiological surveillance sectors of the Departments of the Family Health Strategy Women's Health Program (Figure S1).

Variables such as age group, marital status, education level, and ethnicity, which were available from death certificates and associated with mortality due to uterine neoplasms, were analyzed.

The age group was divided in decades and the marital status was classified in women with partners for those death certificates indicating the women were married and women without partners for those who were single, divorced or widowed. The education level variable was divided in years of study according to Law 5692/71, which considers primary education to be up to 8 years of school. The knowledge of the variable can aid planning processes, management and assessment of public health and school policies. Illiterate people require special approaches for the practices of health promotion, protection and recovery[11].

As for the ethnicity, we chose to classify as white and other, which is justified by the human diversity in the degree of miscegenation, from individual to individual, from class to class, from region to region". Also, indicators for indigenous, yellow and black people must be regarded cautiously since these groups are very small in some states and regions in Brazil. In addition, the samples for the whites are more robust, offering a greater guarantee for usage (Brazil, 2008). Additional variables, considered risk factors, were not analyzed because they do not appear on death certificates.

The information on the family income was raised using the data provided by IBGE (Brazilian Institute of Geography and Statistics) through Demographic Census. The systematization of these indicators is in conformity with the international recommendations and contributes for the understanding of how the demographic, social and economic profiles of the population modify, thus, making it possible to monitor social policies and to disseminate relevant information for Brazilian society. Additional variables, considered risk factors, were not analyzed because they do not appear on death certificates.

Figure S1. Selection of cases of uterine according to ICD-10.

Data Analysis

Coefficients of mortality (CM) for the primary cause of death were calculated employing a numerator of the total deaths from uterine neoplasms and a denominator of the total female population exposed to the risk multiplied by 100,000. Then, for greater detail, CM calculations from uterine neoplasms were carried out by specific grouping and for each age group, organized in ten-year increments.

The data were entered in spreadsheets using Excel^{R} 2010 for Windows for the adequate storage of information. Statistical analyses were conducted descriptively through calculations of the mean, median, maximum and minimum values, standard deviation, and absolute and relative frequencies (percentage), along with line graphs. The inferential analyses employed with the intention of confirming or refuting evidence found in the descriptive analysis was the Pearson chi-square test.

For the trend analysis, a polynomial regression model was utilized in which CM were considered the dependent variable (y), and years of schooling were the independent variable (x). The variable “year” was transformed in a year-centered variable (x-2009), and the series were smoothed through the moving average of three points.

The linear, quadratic, and cubic polynomial regression models were tested. However, after verification of the dispersion diagram, only the first-order (linear) equations were retained for this study. The significance level was set at α <0.05 for all models. For the selection of the best-fitting model, the dispersion diagram, value of the coefficient of determination (R-squared) and analysis of the residuals (assumptions of true homoscedasticity were considered. When all criteria were found to be significant for multiple models and when the coefficients of determination were similar, simpler models were selected.

A choropleth map was constructed to show the distribution of mortality percentages by health center, overlaying the average population income by census tract of the studied municipality using the open-source software QGIS version 2.8 (REF). The socioeconomic income variable was obtained for each census tract through the website of the city health department of Maringá [9]. The descriptive and inferential analyses were conducted in Microsoft Excel (version 2013), Epi Info 2005 and SPSS, version 20.1. The significance level for all analyses was set at α <0.05. Information collected was collected after approval for the Research Ethics Committee of the State University of Maringá with protocol number 86.4889.

Results

Over the period studied, there were a total of 8.924 cases of mortality (44.3%) in women due to neoplasms, 147 (1.65%) of which were specifically uterine neoplasms. Among deaths from uterine neoplasms, 57.8% were malignant uterine neoplasms of unspecified origin.

The mean age of these women was 62 years with a standard deviation of 15 years. In relation to education level, 71.5% of women had completed less than 8 years of formal education. 79.6% of women were white, and 60.5% of women did not have a partner. The results were meaningful regarding the ethnic group for the three types of uterine cancer; as for the marital status, there was significance for ICD C54 and C55, and only for C54 regarding education level. (Table S1)

Variable			C53			C54			C55		
			2004-2008	2009-2014	*p*	2004-2008	2009-2014	*p*	2004-2008	2009-2014	*p*
	Total	%	n (%)	n (%)		n (%)	n (%)		n (%)	n (%)	
Age Group					0.163			0.244			1.000
25-34	5	3.4	0 (0.0)	4 (7.7)		0 (0.0)	0 (0.0)		0 (0.0)	1 (5.3)	
35-44	15	10.2	5 (14.7)	8 (15.4)		0 (0.0)	0 (0.0)		0 (0.0)	1 (5.3)	
45-54	19	12.9	5 (14.7)	8 (15.4)		0 (0.0)	0 (0.0)		1 (5.9)	2 (10.5)	
55-64	37	25.2	6 (17.6)	12 (23.1)		2 (28.6)	5 (27.8)		10 (58.8)	6 (31.6)	
≥ 65	71	48.3	18 (52.9)	20 (38.5)		5 (71.4)	13 (72.2)		6 (35.3)	9 (47.4)	
Education Level					0.059			1.000*			
(years of study)											0.713
< 8	103	71.5	28 (82.4)	33 (63.5)		6 (85.7)	14 (77.8)		11 (64.7)	13 (68.4)	
≥ 8	41	28.5	6 (17.6)	19 (36.5)		1 (14.3)	4 (22.2)		6 (35.3)	6 (31.6)	
Ethnicity					0.031*			1.000*			0.684*
White	117	79.6	31 (91.2)	37 (71.2)		6 (85.7)	14 (77.8)		13 (76.5)	16 (84.2)	
Other	30	20.4	3 (8.8)	15 (28.8)		1 (14.3)	4 (22.2)		4 (23.5)	3 (15.8)	
Had a partner/husband					0.598			0.673*			0.192*
With partner/husband	58	39.5	13 (38.2)	17 (32.7)		3 (42.9)	10 (55.6)		5 (29.4)	10 (52.6)	
Without partner/husband	89	60.5	21 (61.8)	35 (67.3)		4 (57.1)	8 (44.4)		12 (70.6)	9 (47.4)	

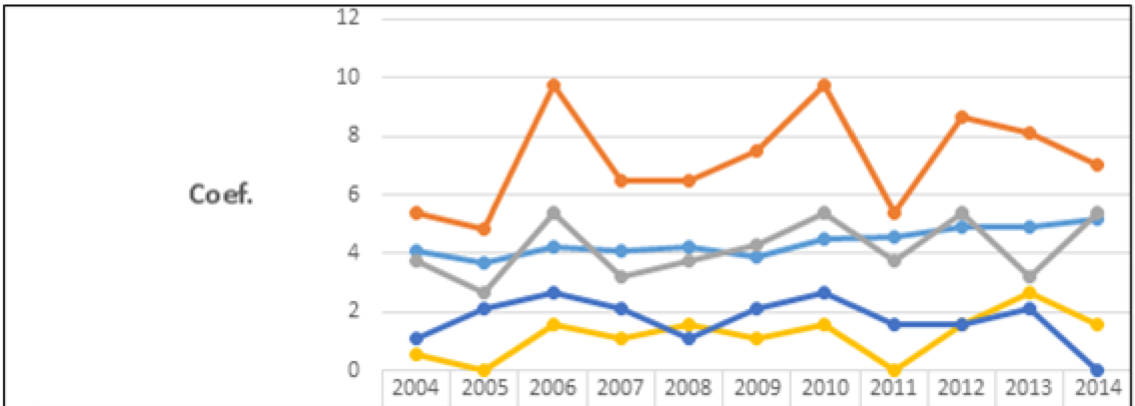
Table S1. The distribution of deaths from uterine neoplasms in women by age group, education level, ethnicity and marital status. Maringá, Paraná, Brazil, 2004-2008 and 2009-2014.

An analysis of the behavior of the CM for uterine neoplasms compared to the CM for general mortality in women revealed greater coefficients for the years of 2006 and 2010. In 2016, the CM for uterine neoplasms and general mortality were 9.7 and 4.2, respectively, whereas these values were 9.7 and 4.5, respectively, in 2010. It is important to note that the CM for uterine neoplasms remained higher than the CM for general mortality in women across all years (Table S2).

Year	Cases of mortality		C53.8	C53.9	C54.1	C54.9	C55	TOTAL	
		Coefficients of mortality (CM)	n	N	n	n	n	N	CM deaths from uterine neoplasms
2004	760	4.1	-	7	1	-	2	10	5.3
2005	688	3.7	-	5	-	-	4	9	4.9
2006	767	4.2	-	10	3	-	5	18	9.7
2007	748	4.1	-	6	2	-	4	12	6.5
2008	778	4.2	-	7	3	-	2	12	6.5
2009	721	3.9	-	8	2	-	4	14	7.6
2010	830	4.5	-	10	3	-	5	18	9.7
2011	848	4.6	-	7	-	-	3	10	5.3
2012	911	4.9	-	10	2	1	3	16	8.7
2013	911	4.9	-	6	5	-	4	15	8.1
2014	962	5.2	1	9	3	-	-	13	7.1
Total	8.924	48.3	1	85	24	1	36	147	79.4

Table S2. Behavior of mortality coefficients for uterine neoplasms and general mortality in women, according to ICD-10. Maringá, Paraná, Brazil, 2004 - 2014.

The CM for general mortality in women showed a rising trend, with a mean of 4.36 and an increase of 0.13 per year (r^2=0.89). Furthermore, there was a rising trend in CM for uterine neoplasms, with a mean of 7.41 and an increase of 0.13 per year (r^2=0.51), along with a mean of 4.41 and an increase of 0.12 per year for malignant cervical neoplasms (Figure S2).



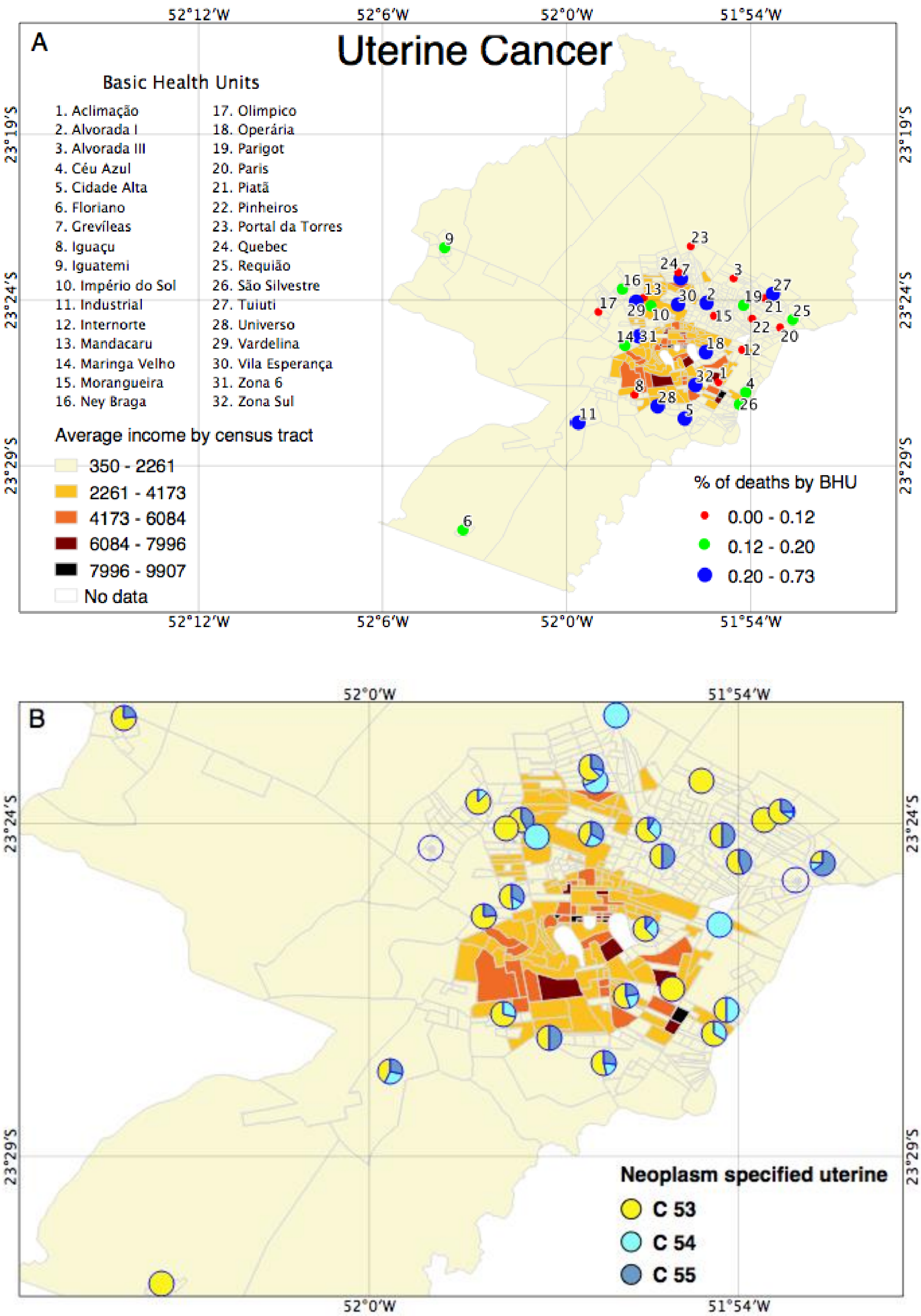
	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
General Mort. Women	4.1	3.7	4.2	4.1	4.2	3.9	4.5	4.6	4.9	4.9	5.2
Uterine Neoplasms	5.39	4.85	9.71	6.47	6.47	7.53	9.71	5.39	8,63	8.09	7.01
C53	3.77	2.69	5.39	3.23	3.77	4.31	5.39	3.77	5.39	3.23	5.39
C54	0.53	0.0	1.61	1.07	1.61	1.07	1.61	0.0	1.61	2.69	1.61
C55	1.07	2.15	2.69	2.15	1.07	2.15	2.69	1.61	1.61	2.15	0.0

	Model	R^2	*p*	Tendency
General Mort. Women	y=4.36+0.13x	0.89	*<0.001*	Rising
Uterine Neoplasms	y=7.41+0.13x	0.51	*0.030*	Rising
C53	y=4.14+0.12x	0.50	0.033	Rising
C54	y=1.23+0.09x	0.46	0.053	Constant
C55	y=1.91-0.07x	0.40	0.067	Constant

Figure S1. A selection of uterine neoplasm cases according to ICD-10. Maringá, Paraná, Brasil, 2004-2014.

Next, verification of the percent distribution of mortality due to uterine neoplasms was sought by the location and population serviced by the referenced health center, to better understand the reality of the population and to target specific actions for the reduction of these coefficients. The study showed that the Vardelina and Grevílea health centers possessed a lower mean household income and a greater mean number of deaths, corresponding to 0.73% and 0.54% of the population, respectively, whereas the mean percentage of deaths in the municipality was 0.15%

In figures S3A and S3B, it is evident that mortality from neoplasms of the cervix (C53) occurs with greater frequency in the majority of health centers and that all neoplasms are more prevalent in areas that have a less favorable value of mean household income.



Figures S3A-B. Percent distribution of deaths from uterine neoplasms in women, by health center, Maringá, Paraná, Brazil, 2004 to 2014.

Discussion

Literature involving data on mortality trends in uterine neoplasms that distinguish between the cervix (C53), uterus (C54), and unspecified origin (C55) is lacking. Therefore, the objective of this study was to present data on mortality trends due to these neoplasms. Mortality attributed to unspecified cervical neoplasms (ICD-10, C53.9) most frequently affected women. The tendency of the CM for uterine neoplasms rose with a mean of 7.41 and an increase of 0.13 per year. There was a mean of 4.14 and an increase of 0.12 per year for cervical neoplasms.

The behavior of age groups in the study revealed an increase in the number of mortality cases in women 65 years or older. This result may reveal the deficiencies of the public health system regarding young women actively going through preventive examinations, since these neoplasms are silent and take a long time to reach the final stage of the disease. Findings of the variable of education level indicated that a majority of women had less than 8 years of formal schooling. Schooling is an indirect indicator of the family's socioeconomic status and a risk predictor of chronic disease. This factor may be related to women not understanding the orientations made by professionals, as well as the malignancy of this pathogenesis. Given this hypothesis, we believe that this variable may be indicative of improvements in the approach strategies in the guidelines.

There was a significant increase among white women in relation to marital status, with a majority of deaths occurring in women without partners. This piece of information may indicate the likely multiplicity of partners, thus being exposed to sexually transmitted diseases, increasing the risk for uterine neoplasms.

The distribution of deaths by location revealed that the Vardelina health center experienced the greatest amount of deaths over the studied period, with a percentage higher than the municipalities, suggesting also that the percentage of deaths was greater in health centers closer to census tracts with a lower mean household income.

In figures S3A and S3B, it is evident that mortality from neoplasms of the cervix (C53) occurs with greater frequency in the majority of health centers and that all neoplasms are more prevalent in areas that have a less favorable value of mean household income. The debate over inequality in income distribution has grown in recent years. In a global perspective, Latin American countries in general, and particularly Brazil, stand out for this inequality, which tends to limit equal opportunities in society and may be an inhibitor of socioeconomic growth [12]. Since one of the risk factors related mainly to cervical cancer is socioeconomic conditions, the research is in line with perspectives related to the economy and health of women in less favored regions.

If, on one hand, assistance technologies have decisively influenced results in the area, with significant advances; on the other hand, they accentuate inequalities in societies where access to this type of assistance is not guaranteed for the whole population, as is the case of Brazil. Small differences in the access to secondary (specialties) and tertiary (treatment) services, which are highly effective, can result in large differences in mortality.

Previous studies of non-Hispanic white women have indicated that in 2008, only three types of cancer were responsible for 10% of mortality related to neoplasms overall. It was observed that approximately 460,000 women died from breast cancer, 140,000 women died from ovarian cancer, and 74,000 women died from endometrial cancer, which was found to be the most common among uterine neoplasms [13]. In the Republic of Korea, the mortality incidence per 100,000 women decreased from 12.7 in 2000 to 11.8 in 2008. However, the age group of 25-29 years experienced an elevated rate in 2011 (6.5) than when compared to 2000 (3.6) [14].

Studies carried out in Kanagawa and Japan show that cervical cancer mortality trends are rising in the 20-29-year and 30-39-year age groups, revealing concern for cervical cancer among Japanese women younger than 50 years [15].

A study of cervical cancer mortality in health regions of the Brazilian state of Pernambuco observed a greater number of cases among non-white women. Women who lacked formal schooling altogether and those with lower education levels (1-3 years of schooling) were at an increased risk of death due to the disease, a fact that was found to be independent of race. It was also found that a greater number of deaths from these cancers occurred in unpartnered women compared to married women [16].

Various interactions between health systems and individual-level factors over time likely contributed to the rates observed in the city of Maringá. Diverse changes in Brazilian health policy and other national, state, and municipal initiatives, particularly with respect to women's health, helped to design and implement strategies for the early detection of uterine neoplasms. Such policy changes may have contributed to improvement in the monitoring of indicators related to women's health. However, it would be beneficial if there were continued improvement in the monitoring of data on the capacity of health professionals in relation to early detection and treatment and tracking efforts of uterine neoplasms, which would result in a reduction of coefficients. The data presented here do not correspond to this expectation. Thus, the mortality rate would likely be higher due to better registry infrastructure. However, the variability among years across diverse areas of the city did not conform to this expectation.

The Family Health Strategy has become essential for providing information pertinent to examinations and performing outreach via community health workers to residences, as well as controlling the frequency with which women complete annual examinations. The participation of the entire health team, including members from both the Family Health Strategy and Nucleus for the Support of Family Health, is critical to attracting women.

One potential limitation of this study was the utilization of secondary data, which may be biased from underreporting of mortality due to uterine neoplasms. As estimating depends closely on mortality information, the better the quality of information on mortality is, the better the estimated information is. Since 2005, there has been an improvement in the information on mortality in Brazil, reflected by the quality of the information obtained from the basic cause of death in the death certificate. The current picture, however, still shows some degree of underreporting and high percentage of classification as "ill-defined causes" in some Federal Units. The estimates presented here are, therefore, reflections of this scenario. Another factor to be considered is the progressive and constant search for the improvement of the quality of information, resulting, year after year, in increased validity and accuracy of the annual estimates. Although there are limitations, the estimates are believed to be capable of describing current patterns of cancer deaths, making it possible to assess the magnitude and impact of this disease in Brazil [3].

Our findings carry implications that support the notion that health efforts should be reconsidered and reformulated, considering greater planning and monitoring strategies. In this context, we emphasize the importance that the issues examined in this study hold for healthcare professionals and stewards alike. These groups need to be capable of identifying risk factors in women and subsequently offering adequate care and coordination to reduce mortality indicators.

Studies that attempt to demonstrate the interrelationship among the SIAB, The Cervical Cancer Information System, and The Family Health Strategy Laboratory System are fundamental to understanding and mapping out specific efforts and program strategies. Research that seeks to associate cervical cancer risk factors and strategies that improve examination coverage are crucial for the reduction of mortality indicators.

Conclusion

This study identified a high number of deaths from uterine neoplasms, primarily uterine neoplasms of an unspecified origin. In doing so, this study also showed that mortality due to uterine neoplasms is increasing annually over the study period, mainly in women who were 65 years or older, had lower education levels, and had low incomes, which confirms past research in Brazil.

These data indicate that those in disadvantaged social strata are those with lower access to health services and a diminished quality of information surrounding primary prevention and the effective treatment of uterine neoplasms.

Our findings carry implications that support the notion that health efforts should be reconsidered and reformulated, taking greater planning and monitoring strategies into account. In this context, we emphasize the importance that the issue approached in this study holds for healthcare professionals and stewards alike. These groups need to be capable to identify risk factors in women and subsequently offer adequate care and coordination in order to reduce mortality indicators.

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Supporting Information

Figure S1. A selection of uterine neoplasm cases according to ICD-10. Maringá, Paraná, Brasil, 2004-2014.

Table S1. The distribution of deaths from uterine neoplasms in women by age group, education level, ethnicity and marital status. Maringá, Paraná, Brazil, 2004-2008 and 2009-2014.

Table S2. Behavior of mortality coefficients for uterine neoplasms and general mortality in women, according to ICD-10. Maringá, Paraná, Brazil, 2004 - 2014

Figure S2. The distribution and trend models of mortality coefficients for uterine neoplasms according to the criteria of ICD-10. Maringá, Paraná, Brazil, 2004 to 2014.

Figures S3A-B. Percent distribution of deaths from uterine neoplasms in women, by health center, Maringá, Paraná, Brazil, 2004 to 2014.